

## SAL 100

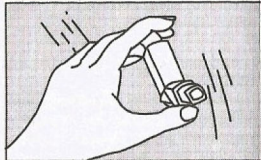
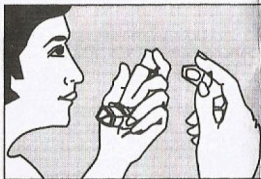
(Salbutamol 100  
MCG/Actuation HFA  
Metered Dose Inhaler)

CFC FREE

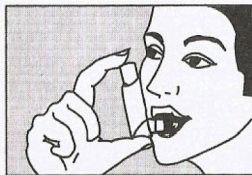
200 Metered Doses



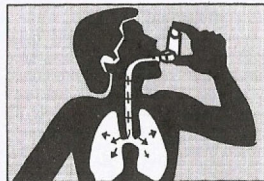
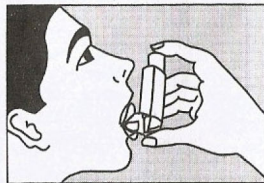
Remove the mouthpiece cover, and check the mouthpiece critically for cleanliness. Shake the inhaler well.



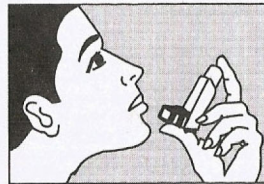
Hold the Inhaler as shown. Breathe out gently through the mouth and immediately place the mouthpiece in the mouth.



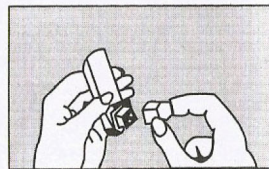
Close the lips around the mouthpiece firmly. Tilt the head slightly backwards. Start breathing in slowly through the mouth. At the same time press the canister as shown, to release one dose while continuing to breathe in steadily and deeply.



Remove the inhaler from the mouth. Hold the breath at least for 10 seconds, or as long as is comfortable. Breathe out slowly.



If another dose is required, wait for at least one minute before repeating steps 1 to 4. After use, replace the mouthpiece cover.



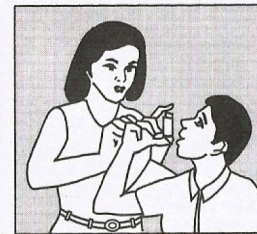
### Note

Do not rush stages 2 and 3. It is important to breathe in slowly through the mouth just before pressing the canister. To be sure of using your inhaler properly, you may initially practice these steps in front of



### For Children

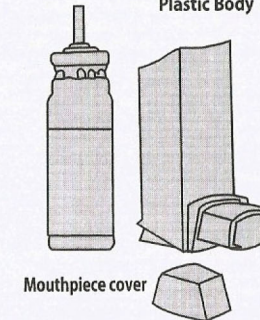
Young children may need help. To help the child, parents must first themselves learn the technique correctly, then properly guide the child to use the inhaler.



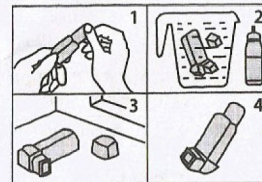
### Parts of Inhaler

Canister

Plastic Body



### Cleansing



### Cleansing :

Clean the inhaler at least once a week.

1. Gently pull the metal canister out of the plastic body of the inhaler. Remove the mouthpiece cover.
2. Rinse the plastic body and the mouthpiece cover in warm water.
3. Leave to dry in warm place. Avoid excessive heat.
4. Replace the canister and the mouthpiece cover correctly.
5. Do not put metal canister in water.



**SALBUTAMOL 100 MCG/ACTUATION  
HFA METERED DOSE INHALER**



## PATIENT'S INFORMATION LEAFLET

Before using the inhaler read this leaflet carefully and follow the instructions.

### SAL 100

(Salbutamol 100  
MCG/Actuation HFA  
Metered Dose Inhaler)

**CFC FREE**

### 200 Metered Doses

#### Description and Composition:

An easily actuated aerosol containing a white to off white suspension with characteristic odor in a suitable liquid, in a suitable pressurized canister fitted with valve.

Each actuation delivers Sulbutamol BP 100mcg.

#### Pharmacodynamics:

Salbutamol stimulates  $\beta$  - adrenergic receptors and has little or no effect on the  $\alpha$ -adrenergic receptors. Salbutamol is relatively selective  $\beta_2$  -adrenergic agonist.  $\beta$  -adrenergic agonists stimulate the production of cyclic adenosine-3',5'- monophosphate (AMP) by activation of the enzyme adenyl cyclase. Cyclic AMP appears to mediate numerous cellular responses. Increased intracellular cyclic AMP enhances the activity of cAMP

dependent protein kinase A, which inhibits the phosphorylation of myosin and lowers intracellular calcium concentrations, resulting in smooth muscle relaxation. Increased intracellular cyclic AMP concentrations also are associated with inhibition of the release of mediators from mast cells in airways.

#### Pharmacokinetics:

##### Absorption :

Salbutamol appears to be absorbed from the respiratory tract over several hours following oral inhalation. The peak plasma salbutamol concentration occurs in 2-5 hours. Metabolism and Elimination: After oral inhalation of salbutamol in healthy adults, the elimination half-life of unchanged drug was determined indirectly to be 3.8 hours. Salbutamol is extensively metabolized in the liver, mainly to salbutamol 4'-O- sulfate which has little or no  $\beta$ -adrenergic stimulating effect and no  $\beta$ -adrenergic blocking effect. Salbutamol and its metabolites are rapidly excreted in urine and faeces. After oral inhalation of salbutamol in patients with asthma, approximately 70% of a dose is excreted in urine as unchanged drug and metabolites within 24 hours and 80-100% within 72 hours; about 30% of the dose is excreted in urine unchanged in 24 hours. About 10% of an inhaled dose of salbutamol may be excreted in faeces.

#### Indications:

Salbutamol inhaler is indicated in the treatment and prophylaxis of bronchospasm in bronchial asthma, chronic bronchitis and emphysema.

#### Side effects:

Mild tremor and headache have been rarely reported. These usually with continued treatment as with other inhalation therapy the potential for paradoxical bronchospasm should be kept in mind. If it occurs, the therapy should be discontinued immediately and alternative therapy instituted. Hypokalemia may also result from  $\beta_2$ -agonist therapy. Hypersensitivity reactions including angioedema, urticaria, bronchospasm, hypotension and collapse have been reported very rarely.

#### Warning and Precautions:

Salbutamol preparations should be used with caution in patients suffering from thyrotoxicosis. Adverse metabolic effect of high doses of salbutamol may be exacerbated by concomitant administration of high doses of corticosteroids; patients should therefore be monitored carefully when the 2 forms of therapy are used together although this precaution is not so applicable to inhalation therapy. Propranolol and other  $\beta$ -adrenoceptor blocking agents antagonize the effects of salbutamol. Hypokalaemia associated with the doses of salbutamol may result in increased susceptibility to digitalis-induced cardiac arrhythmias. The effects of salbutamol, such as hypokalaemia, may also be enhanced by concomitant administration of aminophylline or other xanthines; hypokalaemia can also be enhanced by diuretic therapy.

#### Pregnancy and Lactation:

##### Pregnancy

Administration of the drug during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus.

##### Nursing mothers

As salbutamol is probably secreted in breast milk, its use in nursing mothers requires careful consideration.

#### Contraindications:

Hypersensitivity to any of the components of the formulation.

#### Drug interactions:

Salbutamol and non-selective  $\beta$ -blocking drugs such as propranolol, should not be usually prescribed together. Potentially serious hypokalemia may result from  $\beta_2$  -agonist therapy. Particular caution is advised in acute, severe asthma as this effect may be potentiated by concomitant treatment with xanthine derivatives, steroids, diuretics and by hypoxia. It is recommended that serum potassium levels are monitored in such situations.

#### Dosage and administration:

Adults and children over four years of age

For the relief of acute episodes of bronchospasm or prevention of asthma symptoms; one or two inhalations four to six times daily. If control of mild asthma deteriorates and results in regular use of a short acting  $\beta_2$  agonist (4 times daily), patients should be

instructed to contact a clinician for reevaluation and possible institution of maintenance therapy.

To prevent exercise-induced bronchospasm: Two inhalations 15 minutes prior to exercise.

Children below 4 years of age

Safety and effectiveness of inhaled salbutamol have not been established.

#### Method of administration:

Oral inhalation.

#### Overdosage, treatment and antidotes:

Overdosage of orally inhaled salbutamol produced no fatalities. The commonest symptoms were tremor, flushing, agitation and palpitations due to tachycardia.

Usually no specific treatment is necessary.

In the management of overdose, the manufacturers suggest that a relatively selective  $\beta_1$  adrenergic blocking agent may be used, if necessary, but only with extreme caution in asthmatic patients because an asthmatic attack may be induced. Dialysis is no appropriate therapy for the management of salbutamol overdose.

#### Storage:

Salbutamol inhaler should be stored in a dry place, below 30°C and protect from light. Inhaler should not be refrigerated.

#### Cautions:

The metal canister of salbutamol inhaler is pressurized. It should not be broken, punctured or burnt, even when apparently empty. Shelf life: 3 years from the date of manufacture.

#### Packing/size of pack:

Each pack containing one aerosol container with product label, pack insert and an oral adaptor.

#### Presentation:

Salbutamol Inhaler HFA 100 mcg / actuation available in 200 metered doses.

#### Self Life :

3 years from the date of manufacturer.

Date of Revision: Dec, 2012



Manufactured by:  
**Midac-Care Pharmaceuticals Pvt. Ltd.**  
B-16, MIDC, Waluj, Aurangabad 431 136,  
INDIA